



EEE240

Fundamentals of Digital Logic Design

Chapter no. 3

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Outline

- Introduction to gate-level minimization
- The map method
- Four variable K-Map
- PoS simplification
- Don't care condition

Must Reading

- Chapter No. 3: Gate-Level Minimization

Introduction

“Gate-level minimization is the design task of finding an optimal gate-level implementation of the Boolean functions describing a digital circuit”

- Manually difficult for several inputs
- Logic synthesis tools can do it very efficiently and quickly, BUT
- Designer must understand underlying mathematical description

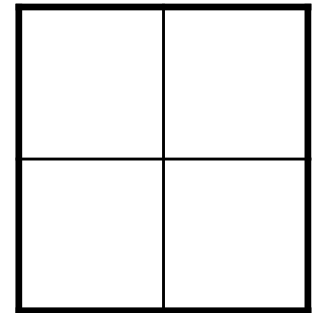
“will enable you to execute a manual design of simple circuits, preparing you for skilled use of modern design tools”

The Map Method

- Truth table is unique, but algebraic expressions can be many
- Rules for simplification of an algebraic expressions are intuitive and not straight forward
- *Karnaugh map or K-map*:
 - simple, straightforward procedure for minimizing Boolean functions
 - Diagram made of square
 - Each square represents one (1) min term
 - Enables visualize all possible ways of expressing a Boolean algebraic function
 - Can give simplest expression
- Simplest expression?
 - Minimum number of terms and with the smallest possible number of literals in each term
 - expression produces a circuit diagram with a minimum number of gates and the minimum number of inputs to each gate

Karnaugh Map

- Adjacent Squares
 - Number of squares = number of combinations
 - Each square represents a minterm
 - 2 Variables $\Rightarrow$ 4 squares
 - 3 Variables $\Rightarrow$ 8 squares
 - 4 Variables $\Rightarrow$ 16 squares
 - Each two adjacent squares differ in one variable
 - Two adjacent minterms can be combined together



Example: $F = x y + x y' = x (y + y') = x$

Note: adjacent squares horizontally and vertically *NOT* diagonally

Two Variable K-Map

- Example

	$x \ y$	F	Minterm	
0	0 0	0	m_0	$\bar{x} \bar{y}$
1	0 1	0	m_1	$\bar{x} y$
2	1 0	0	m_2	$x \bar{y}$
3	1 1	1	m_3	$x y$

m_0	m_1
m_2	m_3

	y
	0
x	0
	1

	y
	0
x	0
	1
0	$\bar{x} \bar{y}$
	$\bar{x} y$
1	$x \bar{y}$
	$x y$

Two Variable K-Map

- Example

	$x \ y$	F	Minterm	
0	0 0	0	m_0	$\bar{x} \bar{y}$
1	0 1	1	m_1	$\bar{x} y$
2	1 0	1	m_2	$x \bar{y}$
3	1 1	1	m_3	$x y$

m_0	m_1
m_2	m_3

	y	
	0	1
x	0	1
	1	1

$$F = \bar{x} y + x y + x \bar{y}$$

$$(\bar{x} + x) y$$

$$x(y + \bar{y})$$

	y	
	0	1
x	0	1
	1	1

Three-variable Map

	x	y	z	Minterm	
0	0	0	0	m_0	$\bar{x} \bar{y} \bar{z}$
1	0	0	1	m_1	$\bar{x} \bar{y} z$
2	0	1	0	m_2	$\bar{x} y \bar{z}$
3	0	1	1	m_3	$\bar{x} y z$
4	1	0	0	m_4	$x \bar{y} \bar{z}$
5	1	0	1	m_5	$x \bar{y} z$
6	1	1	0	m_6	$x y \bar{z}$
7	1	1	1	m_7	

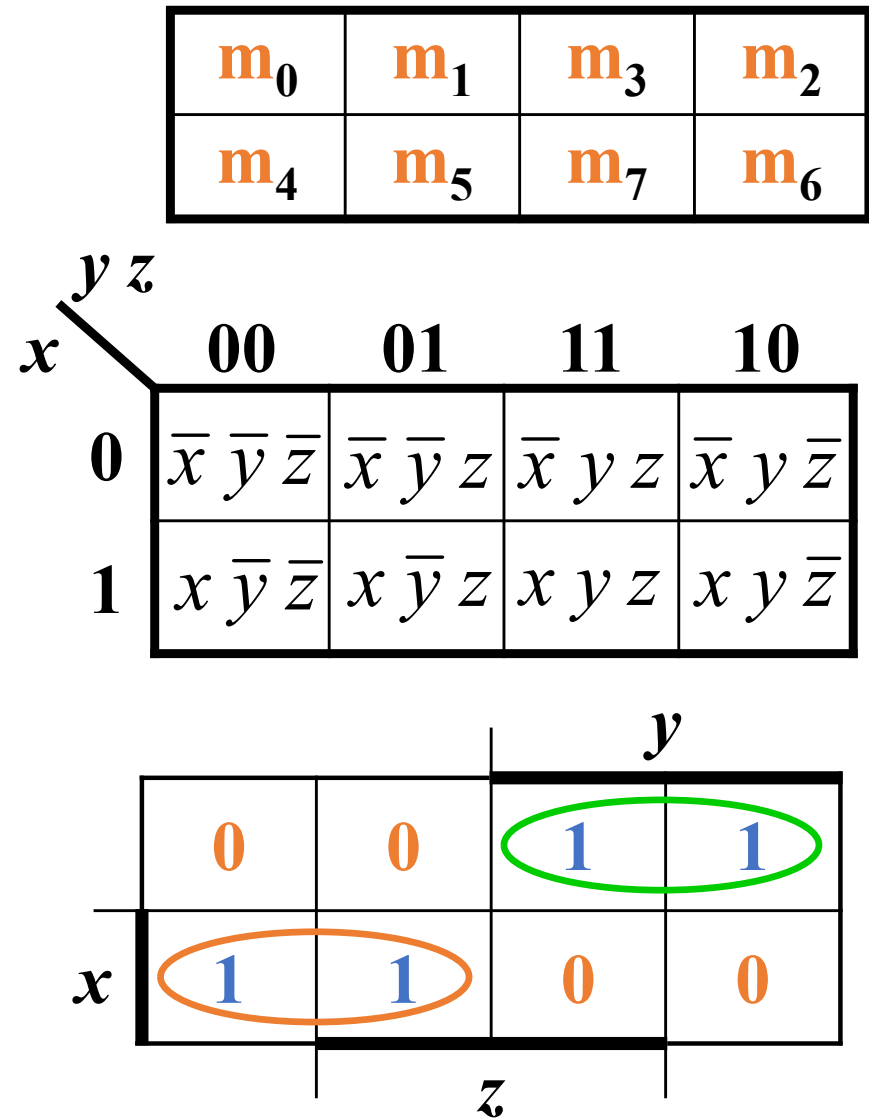
m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6

		yz			
		00	01	11	10
x	0	$\bar{x} \bar{y} \bar{z}$	$\bar{x} \bar{y} z$	$\bar{x} y z$	$\bar{x} y \bar{z}$
	1	$x \bar{y} \bar{z}$	$x \bar{y} z$	$x y z$	$x y \bar{z}$

Three-variable Map

- Example

	x y z	F	Minterm	
0	0 0 0	0	m_0	$\bar{x} \bar{y} \bar{z}$
1	0 0 1	0	m_1	$\bar{x} \bar{y} z$
2	0 1 0	1	m_2	$\bar{x} y \bar{z}$
3	0 1 1	1	m_3	$\bar{x} y z$
4	1 0 0	1	m_4	$x \bar{y} \bar{z}$
5	1 0 1	1	m_5	$x \bar{y} z$
6	1 1 0	0	m_6	$x y \bar{z}$
7	1 1 1	0	m_7	$x y z$



$$F = x \bar{y} + \bar{x} y$$

Three-variable Map

- Example

	x	y	z	F	Minterm
0	0	0	0	0	m_0 $\bar{x} \bar{y} \bar{z}$
1	0	0	1	0	m_1 $\bar{x} \bar{y} z$
2	0	1	0	0	m_2 $\bar{x} y \bar{z}$
3	0	1	1	1	m_3 $\bar{x} y z$
4	1	0	0	1	m_4 $x \bar{y} \bar{z}$
5	1	0	1	0	m_5 $x \bar{y} z$
6	1	1	0	1	m_6 $x y \bar{z}$
7	1	1	1	1	m_7 $x y z$

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6

		yz			
		00	01	11	10
x	0	$\bar{x} \bar{y} \bar{z}$	$\bar{x} \bar{y} z$	$\bar{x} y z$	$\bar{x} y \bar{z}$
	1	$x \bar{y} \bar{z}$	$x \bar{y} z$	$x y z$	$x y \bar{z}$

		y			
		0	0	1	0
x	0	0	0	1	0
	1	1	0	1	1
		z			

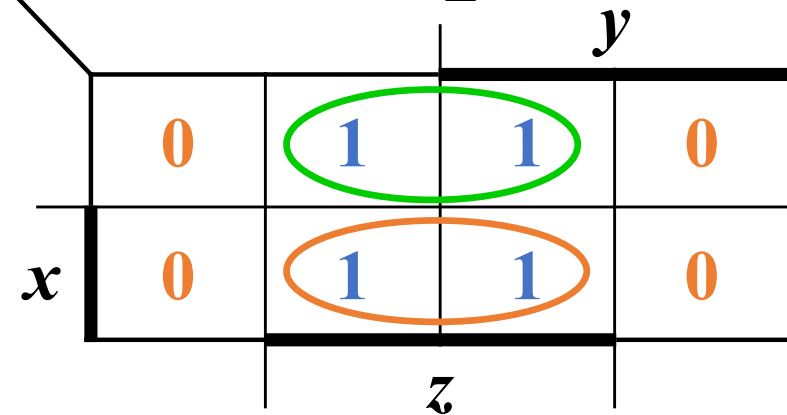
Extra

$$F = \bar{x} \bar{z} + yz + \cancel{xy}$$

Three-variable Map

- Example

	x	y	z	F	Minterm
0	0	0	0	0	m_0 $\bar{x} \bar{y} \bar{z}$
1	0	0	1	1	m_1 $\bar{x} \bar{y} z$
2	0	1	0	0	m_2 $\bar{x} y \bar{z}$
3	0	1	1	1	m_3 $\bar{x} y z$
4	1	0	0	0	m_4 $x \bar{y} \bar{z}$
5	1	0	1	1	m_5 $x \bar{y} z$
6	1	1	0	0	m_6 $x y \bar{z}$
7	1	1	1	1	m_7 $x y z$



$$F = \underbrace{\bar{x} \bar{y} z + \bar{x} y z}_{\bar{x} z (\bar{y} + y)} + \underbrace{x \bar{y} z + x y z}_{x z (\bar{y} + y)}$$

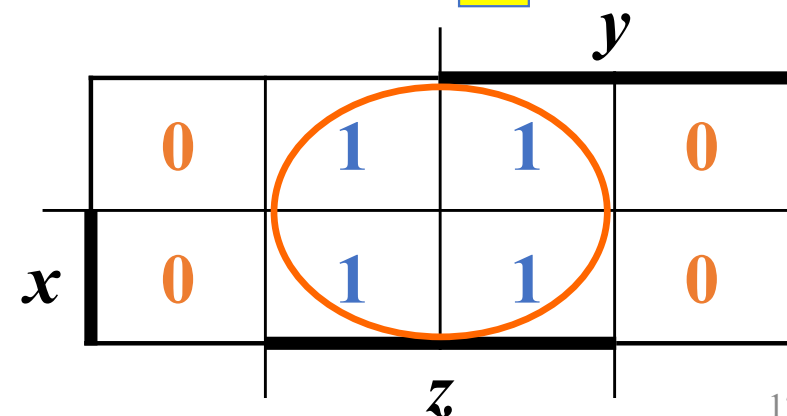
$$\bar{x} z (\bar{y} + y)$$

$$x z (\bar{y} + y)$$

$$\bar{x} z$$

$$x z$$

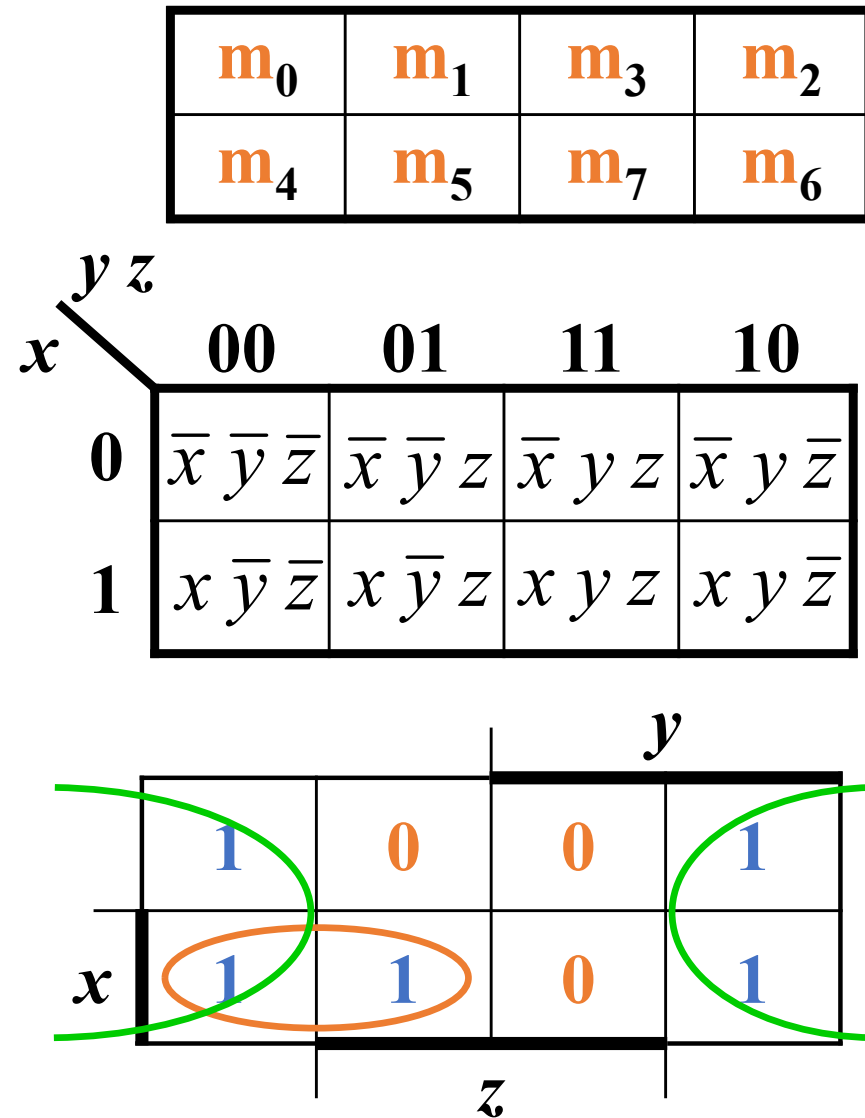
$$z$$



Three-variable Map

- Example

	x y z	F	Minterm	
0	0 0 0	1	m_0	$\bar{x} \bar{y} \bar{z}$
1	0 0 1	0	m_1	$\bar{x} \bar{y} z$
2	0 1 0	1	m_2	$\bar{x} y \bar{z}$
3	0 1 1	0	m_3	$\bar{x} y z$
4	1 0 0	1	m_4	$x \bar{y} \bar{z}$
5	1 0 1	1	m_5	$x \bar{y} z$
6	1 1 0	1	m_6	$x y \bar{z}$
7	1 1 1	0	m_7	$x y z$



$$F = \bar{z} + x \bar{y}$$

Four-variable Map

	w x y z	Minterm	
0	0 0 0 0	m_0	$\bar{w} \bar{x} \bar{y} \bar{z}$
1	0 0 0 1	m_1	$\bar{w} \bar{x} \bar{y} z$
2	0 0 1 0	m_2	$\bar{w} \bar{x} y \bar{z}$
3	0 0 1 1	m_3	$\bar{w} \bar{x} y z$
4	0 1 0 0	m_4	$\bar{w} x \bar{y} \bar{z}$
5	0 1 0 1	m_5	$\bar{w} x \bar{y} z$
6	0 1 1 0	m_6	$\bar{w} x y \bar{z}$
7	0 1 1 1	m_7	$\bar{w} x y z$
8	1 0 0 0	m_8	$w \bar{x} \bar{y} \bar{z}$
9	1 0 0 1	m_9	$w \bar{x} \bar{y} z$
10	1 0 1 0	m_{10}	$w \bar{x} y \bar{z}$
11	1 0 1 1	m_{11}	$w \bar{x} y z$
12	1 1 0 0	m_{12}	$w x \bar{y} \bar{z}$
13	1 1 0 1	m_{13}	$w x \bar{y} z$
14	1 1 1 0	m_{14}	$w x y \bar{z}$
15	1 1 1 1	m_{15}	$w x y z$

m_0	m_1	m_3	m_2
m_4	m_5	m_7	m_6
m_{12}	m_{13}	m_{15}	m_{14}
m_8	m_9	m_{11}	m_{10}

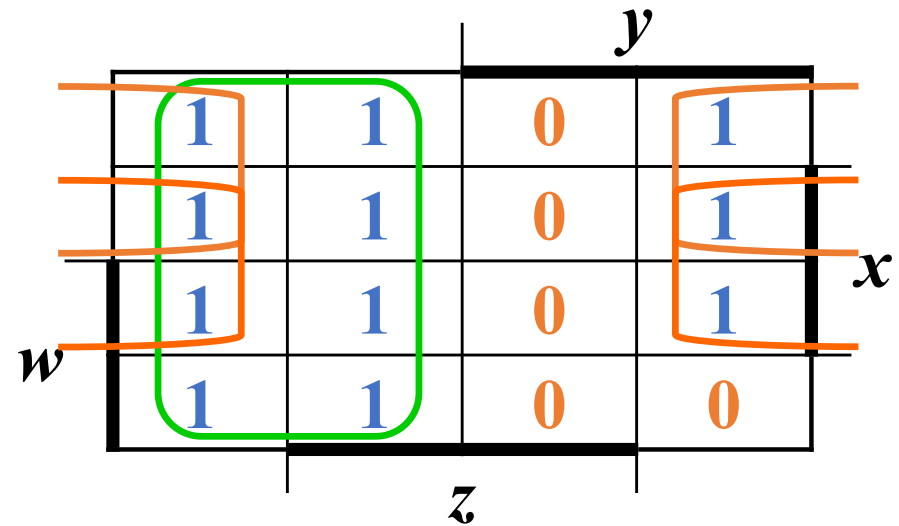
		$y z$			
		00	01	11	10
$w x$	00	$\bar{w} \bar{x} \bar{y} \bar{z}$	$\bar{w} \bar{x} \bar{y} z$	$\bar{w} \bar{x} y \bar{z}$	$\bar{w} \bar{x} y z$
	01	$\bar{w} x \bar{y} \bar{z}$	$\bar{w} x \bar{y} z$	$\bar{w} x y \bar{z}$	$\bar{w} x y z$
	11	$w x \bar{y} \bar{z}$	$w x \bar{y} z$	$w x y \bar{z}$	$w x y z$
	10	$w \bar{x} \bar{y} \bar{z}$	$w \bar{x} \bar{y} z$	$w \bar{x} y \bar{z}$	$w \bar{x} y z$

Four-variable Map

- Example

	w	x	y	z	F	Minterm	
0	0	0	0	0	1	m_0	$\bar{w}\bar{x}\bar{y}\bar{z}$
1	0	0	0	1	1	m_1	$\bar{w}\bar{x}\bar{y}z$
2	0	0	1	0	1	m_2	$\bar{w}\bar{x}y\bar{z}$
3	0	0	1	1	0	m_3	$\bar{w}\bar{x}yz$
4	0	1	0	0	1	m_4	$\bar{w}x\bar{y}\bar{z}$
5	0	1	0	1	1	m_5	$\bar{w}x\bar{y}z$
6	0	1	1	0	1	m_6	$\bar{w}xy\bar{z}$
7	0	1	1	1	0	m_7	$\bar{w}xyz$
8	1	0	0	0	1	m_8	$w\bar{x}\bar{y}\bar{z}$
9	1	0	0	1	1	m_9	$w\bar{x}\bar{y}z$
10	1	0	1	0	0	m_{10}	$w\bar{x}y\bar{z}$
11	1	0	1	1	0	m_{11}	$w\bar{x}yz$
12	1	1	0	0	1	m_{12}	$wx\bar{y}\bar{z}$
13	1	1	0	1	1	m_{13}	$wx\bar{y}z$
14	1	1	1	0	1	m_{14}	$wxy\bar{z}$
15	1	1	1	1	0	m_{15}	$wxyz$

$wx \backslash yz$	00	01	11	10
00	$\bar{w}\bar{x}\bar{y}\bar{z}$	$\bar{w}\bar{x}\bar{y}z$	$\bar{w}\bar{x}y\bar{z}$	$\bar{w}\bar{x}yz$
01	$\bar{w}x\bar{y}\bar{z}$	$\bar{w}x\bar{y}z$	$\bar{w}xy\bar{z}$	$\bar{w}xyz$
11	$wx\bar{y}\bar{z}$	$wx\bar{y}z$	$wxy\bar{z}$	$wxyz$
10	$w\bar{x}\bar{y}\bar{z}$	$w\bar{x}\bar{y}z$	$w\bar{x}y\bar{z}$	$w\bar{x}yz$

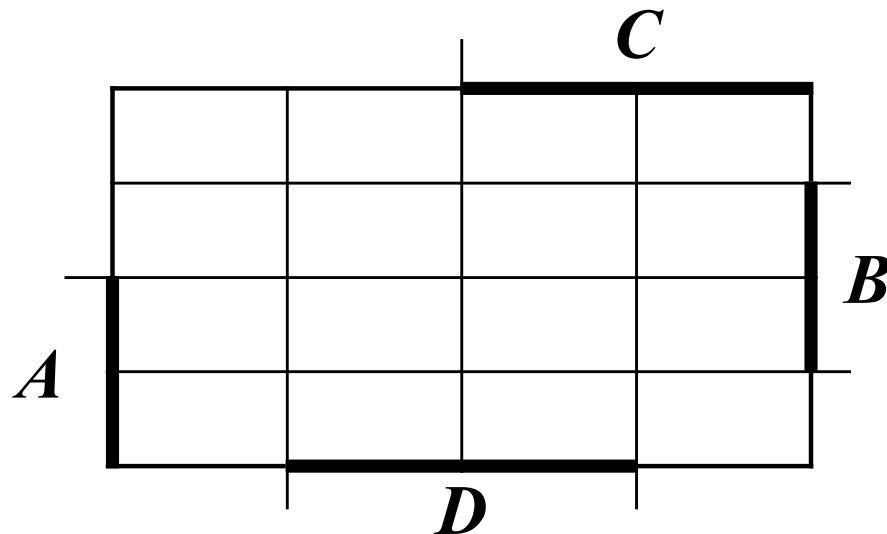


$$F = \bar{y} + \bar{w}\bar{z} + x\bar{z}$$

Four-variable Map

- Example

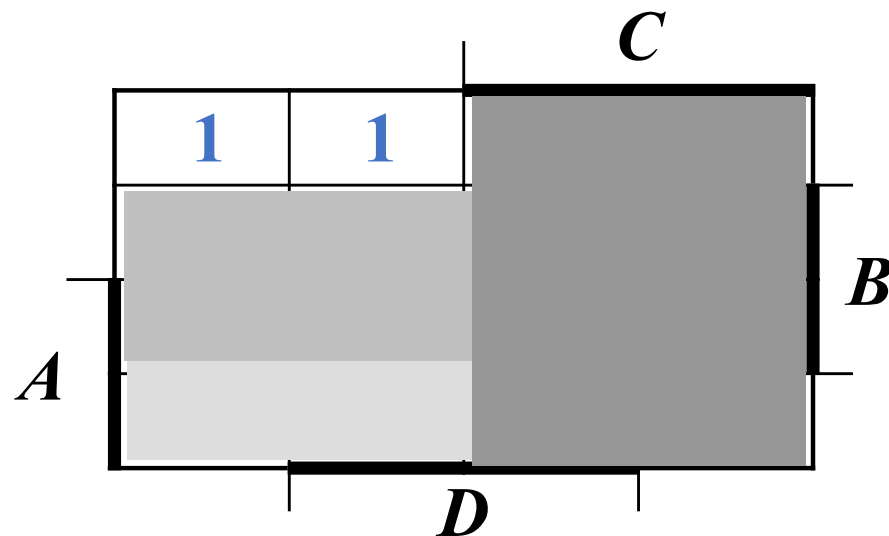
Simplify: $F = A'B'C' + B'CD' + A'BCD' + AB'C'$



Four-variable Map

- Example

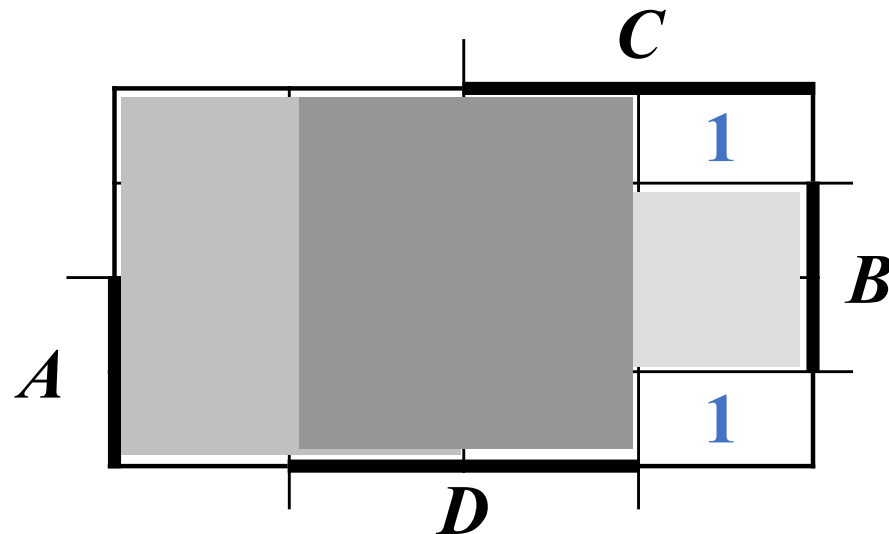
Simplify: $F = A'B'C' + B'CD' + A'BCD' + AB'C'$



Four-variable Map

- Example

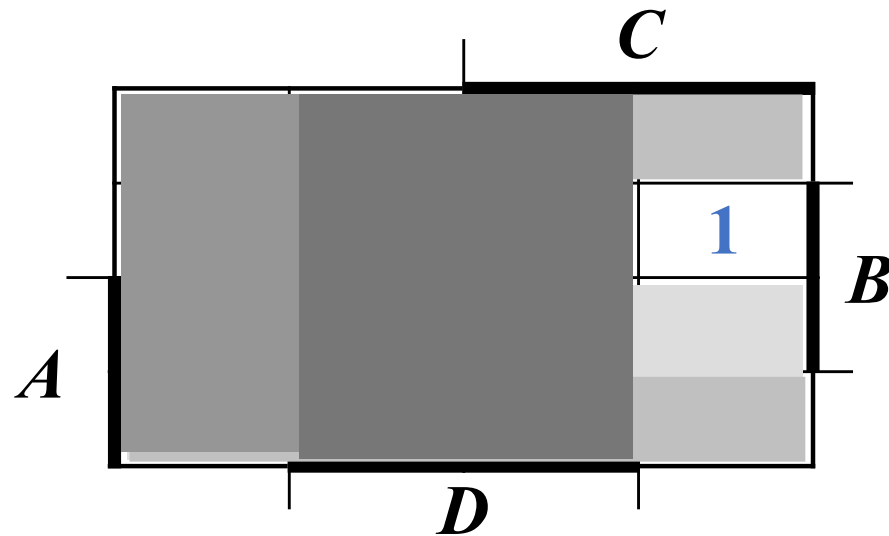
Simplify: $F = A'B'C' + B'CD' + A'BCD' + AB'C'$



Four-variable Map

- Example

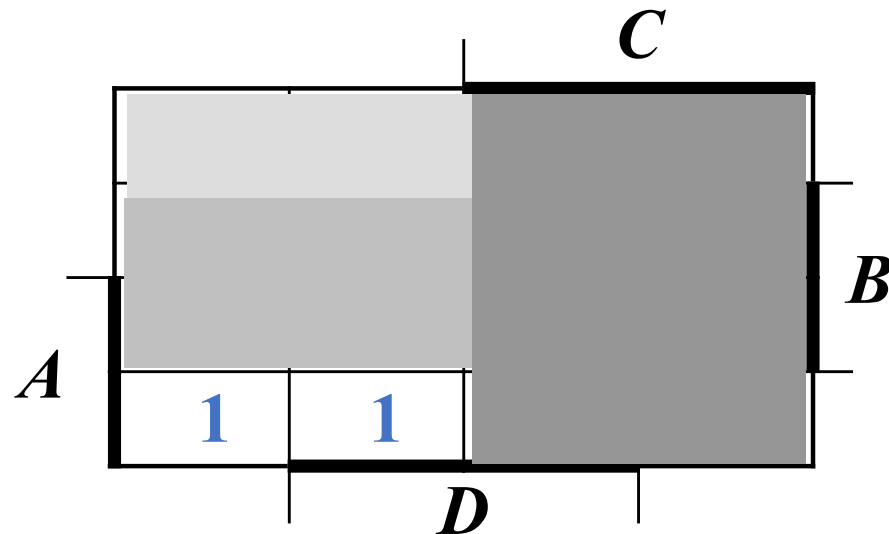
Simplify: $F = A'B'C' + B'CD' + A'BCD' + AB'C'$



Four-variable Map

- Example

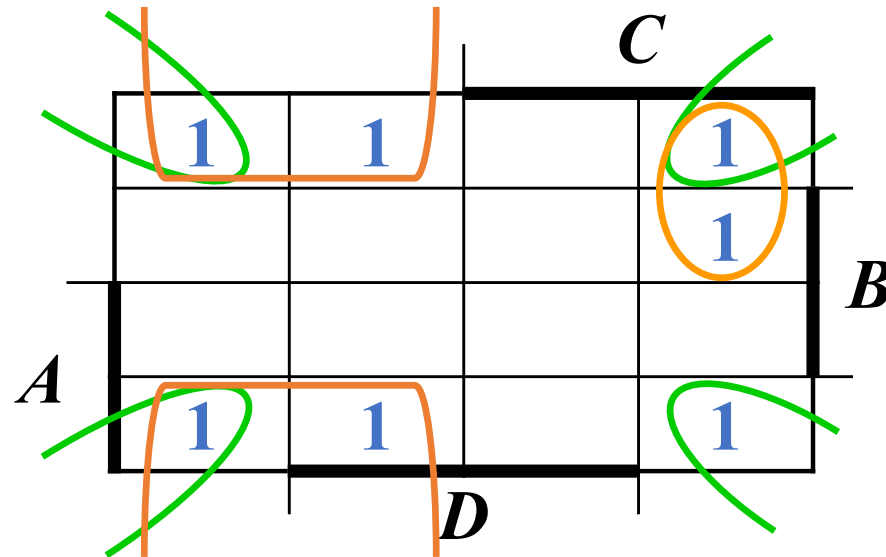
Simplify: $F = A'B'C' + B'CD' + A'BCD' + AB'C'$



Four-variable Map

- Example

Simplify: $F = A' B' C' + B' C D' + A' B C D' + A B' C'$



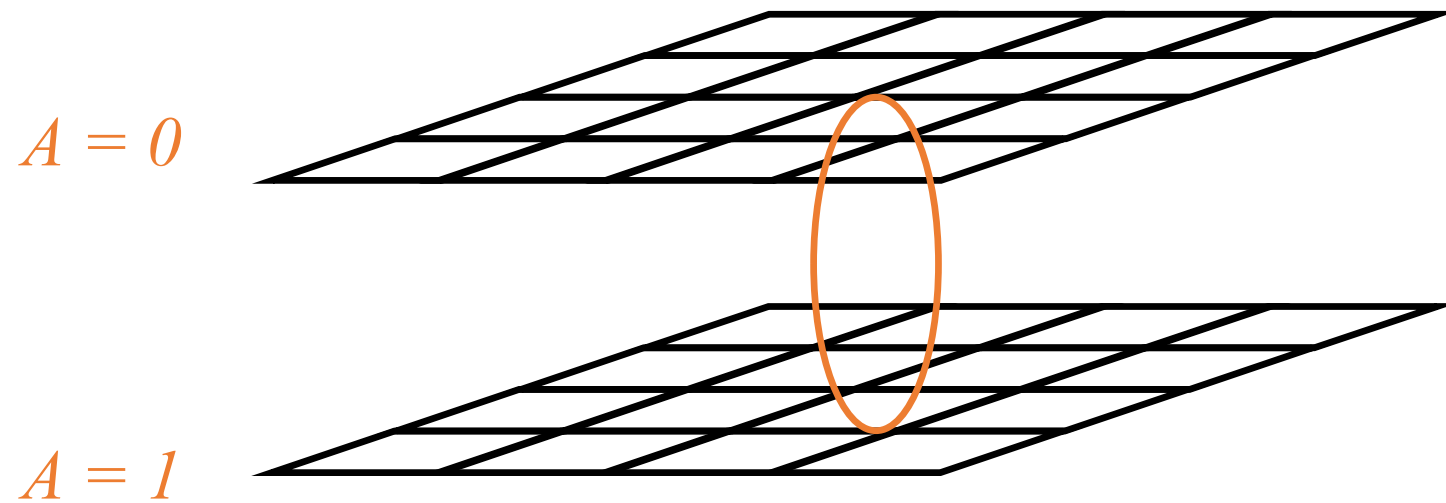
$$F = \overline{B}\overline{D} + \overline{B}\overline{C} + \overline{A}C\overline{D}$$

Five-variable Map

DE		D					
		00	01	11	10		
BC	00	m_0	m_1	m_3	m_2	C	
	01	m_4	m_5	m_7	m_6		
	11	m_{12}	m_{13}	m_{15}	m_{14}		
	10	m_8	m_9	m_{11}	m_{10}		
		E				B	
$A = 0$							

DE		D					
		00	01	11	10		
BC	00	m_{16}	m_{17}	m_{19}	m_{18}	C	
	01	m_{20}	m_{21}	m_{23}	m_{22}		
	11	m_{28}	m_{29}	m_{31}	m_{30}		
	10	m_{24}	m_{25}	m_{27}	m_{26}		
		E				B	
$A = 1$							

Five-variable Map



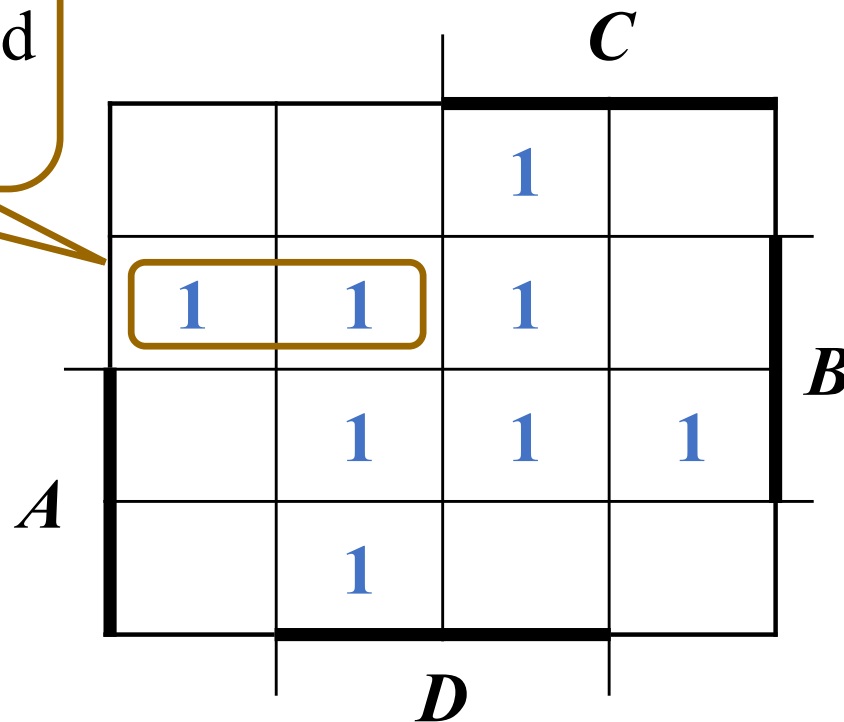
Implicants

Implicant:
Gives $F = 1$

		<i>C</i>	
		1	
	1	1	
<i>A</i>		1	1
	1		
			<i>B</i>
		<i>D</i>	

Prime Implicants

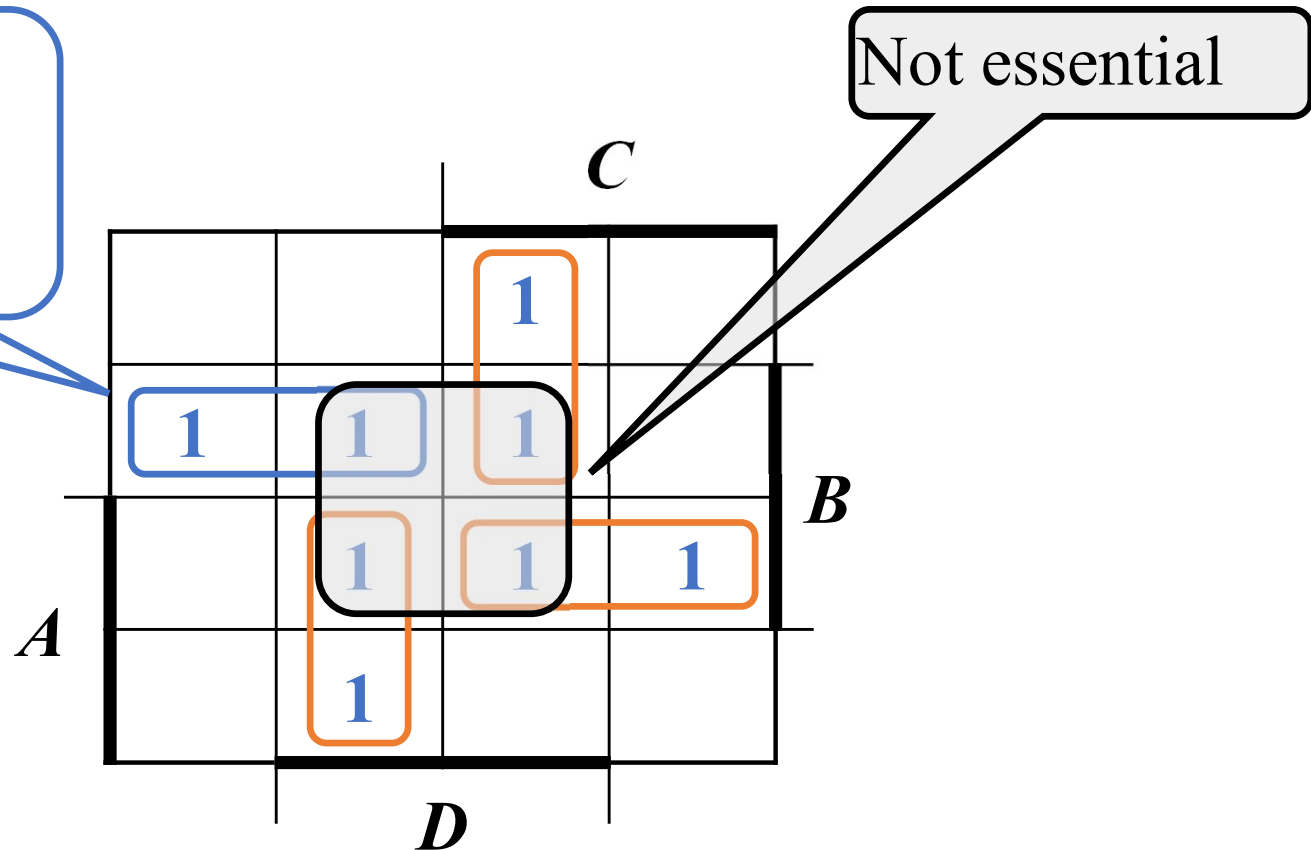
Prime Implicant:
Can't grow beyond
this size



Essential Prime Implicants

8 Implicants, 5 Prime implicants, 4 Essential prime implicants

Essential Prime
Implicant:
No other choice



To ensure that a minimum solution is found, select essential prime implicants first. Then find a minimum set of prime implicants that cover the remaining 1's on the map.